

John D. Soltis

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RESEARCH POSITIONS:

Lasker Data Science Fellow <i>Postdoctoral Prize Fellow</i> Space Telescope Science Institute	2025 - Present	Baltimore, MD
Pre-Doctoral Fellow <i>Advisor: Lehman Garrison</i> Center for Computational Astrophysics, Flatiron Institute	2023 - 2024	New York, NY
PhD Student <i>Advisor: Michelle Ntampaka</i> Johns Hopkins University	2019 - 2025	Baltimore, MD
Affiliate <i>Advisor: Carlton Pennypacker</i> Lawrence Berkeley National Laboratory	2018 - 2019	Berkeley, CA
Research Assistant <i>Advisor: Dragan Huterer</i> University of Michigan	2015 - 2018	Ann Arbor, MI
Summer Scholar <i>Advisors: Jason Chou & John Heebner</i> Lawrence Livermore National Laboratory	2015	Livermore, CA

EDUCATION:

Johns Hopkins University Krieger School of Arts & Sciences Ph.D in Astronomy and Astrophysics	2019 - 2025	Baltimore, MD
Johns Hopkins University Krieger School of Arts & Sciences M.A. in Physics	2019 - 2023	Baltimore, MD
University of Michigan Honors Program, College of Literature, Science, and the Arts B.S. in Physics and Mathematical Physics	2014 - 2018	Ann Arbor, MI

SKILLS & INTERESTS:

General

Machine Learning, Uncertainty Quantification, AI Interpretability,
Hierarchical Bayesian Modeling, Data Analysis, Cosmology, Milky Way Astronomy

Software Packages & Languages

Python, Pytorch, Tensorflow, Unix, LaTeX, MATLAB, Microsoft Excel

FIRST AUTHOR PUBLICATIONS:

1. **J. Soltis**, M. Ntampaka, B. Diemer, J. ZuHone, S. Bose, A. M. Delgado, B. Hadzhiyska, C. Hernández-Aguayo, D. Nagai, H. Trac. “A Multi-Wavelength Technique for Estimating Galaxy Cluster Mass Accretion Rates”, *The Astrophysical Journal*, 985, 2, 212, 12 pp. (2025)
2. **J. Soltis**, L. Garrison. “Self-Similar Mass Accretion Histories in Scale-Free Simulations”, *Monthly Notices of the Royal Astronomical Society*, 532, 2, 1729-1743, (2024)
3. **J. Soltis**, M. Ntampaka, J. Wu, J. ZuHone, A. Evrard, A. Farahi, M. Ho, D. Nagai. “A Machine Learning Approach to Enhancing eROSITA Observations”, *The Astrophysical Journal*, 940, 1, 60, 17 pp. (2022)
4. **J. Soltis**, S. Casertano, A. G. Riess. “The Parallax of Omega Centauri Measured from Gaia EDR3 and a Direct, Geometric Calibration of the Tip of the Red Giant Branch and the Hubble Constant”, *The Astrophysical Journal Letters*, 908, L5 (2021)
5. **J. Soltis**, A. Farahi, D. Huterer, C. M. Liberato. “Percent-Level Test of Isotropic Expansion Using Type Ia Supernovae”, *Phys. Rev. Lett.*, 122, 091301 (2019)

CO-AUTHOR PUBLICATIONS:

1. A. M. Delgado, M. Ntampaka, S. Bose, F. Ferlito, B. Hadzhiyska, L. Hernquist, **J. Soltis**, J. F. Wu, M. Yunus, J. ZuHone. “Estimating the Triaxiality of Massive Clusters from 2D Observables in MillenniumTNG with Machine Learning”, *The Astrophysical Journal*, 1002, 2, 142, 23 pp. (2026)
2. M. Ntampaka, A. Ciprijanovic, A. M. Delgado, **J. Soltis**, J. F. Wu, M. Yunus, J. ZuHone. “The Importance of Being Adaptable: An Exploration of the Power and Limitations of Domain Adaptation for Simulation-Based Inference with Galaxy Clusters”, *arXiv e-prints*, arXiv:2510.09748, (2025)
3. B. E. M. Davis, M. Razavi-Mohseni, **J. Soltis**, H. N. Zhang, E. Kavanagh. “International STEM Graduate Students: A Key to Strengthening the American Economy and Building Competitiveness”, *Journal of Science Policy and Governance*, 25, 1, (2024)
4. M. Ho, **J. Soltis**, A. Farahi, D. Nagai, A. Evrard, M. Ntampaka. “Benchmarks and Explanations for Deep Learning Estimates of X-ray Galaxy Cluster Masses”, *Monthly Notices of the Royal Astronomical Society*, 524, 3, 3289-3302, (2023)
5. W.C. Wan, G. Malamud, A. Shimony, C.A. Di Stefano, M.R. Trantham, S.R. Klein, **J.D. Soltis**, D. Shvarts, R.P. Drake, C.C. Kuran. “Impact of ablator thickness and laser drive duration on a platform for supersonic, shockwave-driven hydrodynamic instability experiments”, *High Energy Density Physics*, 22, (2017)

SELECTED TALKS AND POSTERS:

1. **J. Soltis**, M. Ntampaka. “Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning”, American Astronomical Society Meeting #245, id. 412.02D. *Bulletin of the American Astronomical Society*, Vol. 57, No. 2 e-id 2025n2i412p02 (2025)
2. *Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning*. Cosmology and Galaxy Astrophysics with Simulations and Machine Learning 2024 Conference, Flatiron Institute. December 9th, 2024.
3. *Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning*. Galaxy Cluster Group Meeting, Center for Astrophysics, Harvard-Smithsonian. October 22nd, 2024.
4. *Direct Estimation of Galaxy Cluster Mass Accretion Rates using Machine Learning*. Yale Data Science X Astronomy - Astrophysics Seminar, Yale University. September 17th, 2024.

5. *Estimating Galaxy Cluster Mass Accretion Rates from Observations using Machine Learning*. 2024 AstroAI Workshop, Center for Astrophysics, Harvard-Smithsonian. June 17-21, 2024.
6. *Galaxy Cluster Dynamical State, Follow-Up Observations, and Machine Learning*. Machine Learning Seminar, Argelander Institute for Astronomy, Bonn. June 6th, 2024.
7. *Galaxy Cluster Mass Accretion History*. 6th Neighborhood Workshop, Pennsylvania State University. April 25th, 2024.
8. *Testing the Robustness of Mass Accretion Histories in Scale-Free Simulations*. Merging Clusters Workshop, Yonsei University. December 21st, 2023.
9. **J. Soltis**, M. Ntampaka. “Predicting Follow-Up Observations of Galaxy Clusters Using Machine Learning”, American Astronomical Society Meeting #240, id. 139.19. *Bulletin of the American Astronomical Society*, Vol. 54, No. 6 e-id 2022n6i139p19 (2022)

RESEARCH PROJECTS:

AI Applications in Milky Way Astronomy 2025 - Present
Space Telescope Science Institute
 Training generative AI foundation model on *Gaia* satellite observations of more than 1 billion Milky Way and nearby stars.

Deep Learning Applications in Galaxy Cluster Cosmology 2021 - Present
Johns Hopkins University & Space Telescope Science Institute *PhD Advisor: Michelle Ntampaka*
 Using convolutional neural networks, simulation-based inference, and machine learning interpretability methods to characterize galaxy cluster properties and improve observations.

Robustness of Cosmological Simulations 2023 - 2024
Flatiron Institute *Advisor: Lehman Garrison*
 Investigated the robustness of dark matter halo mass accretion rates in cosmological simulations.

Tip of Red Giant Branch Calibration 2019 - 2020
Johns Hopkins University & Space Telescope Science Institute *Advisor: Adam Riess*
 Measured the Hubble constant using updated position data of Milky Way stars.

Machine Learning Applications in Wildfire Detection 2018 - 2019
Lawrence Berkeley National Laboratory *Advisor: Carl Pennypacker*
 Used images from wildfire detection cameras in California and Nevada to train a convolutional neural network to detect wildfires early.

Testing Statistical Isotropy with Type Ia Supernovae 2017 - 2018
University of Michigan *Advisor: Dragan Huterer*
 Implemented a novel and robust test of statistical isotropy in the Universe using type Ia supernovae residuals.

Simulation of Laser-Driven Plasma Instabilities 2015 - 2016
University of Michigan *Advisor: Matthew Trantham*
 Simulated laser-driven plasma instabilities with a variety of experimental conditions. Results were used to improve experimental design.

National Ignition Facility Support 2015
Lawrence Livermore National Laboratory *Advisors: John Heebner & Jason Chou*
 Improved the accuracy of laser waveform generation in order to aid experiments at the National Ignition Facility.

EXTRACURRICULARS:

Science Policy and Diplomacy Group Congressional Visit Day	2026
Helped organize and train groups for Congressional Visit Day. On the Hill, advocated for the Protecting Free Vaccines Act of 2025, which would preserve vaccine coverage and access.	
Science Policy and Diplomacy Group Maryland Assembly Visit Day	2026
Helped organize and train the groups for meeting with members of the Maryland Assembly. Advocated for oversight of the Human-Relevant Research Fund and MD Law Chapter 448.	
Advocacy Training Seminar	2025
Conducted an informational session for graduate students at Johns Hopkins related to Federal science policy and the role of scientists in advocacy.	
Science Policy and Diplomacy Group Congressional Visit Day	2025
Advocated for robust science funding and against severe budget cuts	
Science Policy and Consulting Career Panel	2025
Hosted five person panel on science policy careers for Johns Hopkins graduate students	
Institute of Electrical and Electronics Engineers Congressional Visit Day	2025
Advocated for science funding, small business funding, and improved STEM workforce policies	
American Astronomical Society Congressional Visit Day	2024
Met with Members of Congress and their staffs to advocate for increased NASA and NSF funding	
Institute of Electrical and Electronics Engineers Congressional Visit Day	2024
Advocated for NSF funding and improved STEM workforce policies	
Science Policy and Diplomacy Group Congressional Visit Day	2024
Advocated for the Keep STEM Talent Act of 2023	
Graduate Representative Organization	2021 - 2023
General Council Representative for the Physics & Astronomy Department	
Maryland Space Grant Observatory Fellow	2021 - 2022
Hosted open house events, trained perspective observers on the telescope, and helped run the MDSGC symposium.	
Public Forum Debate Judge	2020 - 2022
Judged for Michigan Interscholastic Forensics Association Public Forum League	
Michigan Journal of International Affairs	2014 - 2016
Writer for the Asia Region	
Brother Rice Debate Team	2010 - 2014
Varsity in Public Forum Debate	